

Alberto De Luca

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Education

ETH Zürich: *MSc in Robotics, Systems and Control* *Sept 2024 – Current*

- **Relevant Coursework:** Model Predictive Control, Probabilistic AI, Distributed Optimization
- **GPA:** 5.4/6

Politecnico di Milano: *BSc in Automation Engineering* *Sept 2021 – July 2024*

- **Relevant Coursework:** Control Theory, Foundations of Robotics, Circuit Theory, Electrical Machines and Drives
- **Final Grade:** 110 Cum Laude

Experience

Research Intern, ABB Germany *March 2026 – Present*

- Developed a simulation and stability analysis framework in Julia for converter-based power grids, incorporating both AC and DC models.
- Developed a differentiable optimization-based tuning algorithm to enhance grid passivity through controller parameter design.
- Validated the proposed approach on realistic converter models over arbitrary frequency ranges of interest.

Control Engineer, SAPIENS Robotic Hand Prosthesis *April 2023 – July 2024*

- Student-led project of the Automation Engineering Association at Polimi
- Developed real-time control software for microcontrollers in C++, and performed system modeling and simulation in MATLAB/Simulink.

Industrial Intern, Ortoverde *Aug 2021 – Sept 2021*

- Collaborated with the head engineer to collect, process, and analyze production data.
- Supported machine adjustments based on data analysis, reducing material waste by 10%

Selected Projects

Meta Learning Model Predictive Control, Automatic Control Lab

- Semester Project: Developed meta-learning algorithms for Energy Hub MPC to adapt to changing energy prices.
- Built a Python simulation framework using CVXPY and JAX to demonstrate the effectiveness of the proposed method.

Federated Learning for Financial Forecasting

- Research on **privacy preserving** Machine Learning applied to financial markets
- **Achieved:** publishable results; paper accepted at the Computing Conference, London, July 2026. [arxiv](#) [🔗](#)

Robust Truck Thermal Control

- Implemented advanced MPC schemes for thermal regulation in trucks in MATLAB.

System ID and Control Quanser Qube Servo 2

- Performed system identification and control design for an electromechanical servo system using MATLAB/Simulink. [link](#) [🔗](#)

Technical Skills

Programming: Python, C++, MATLAB, Simulink, Bash, Julia

Robotics & Control: State estimation, MPC, acausal modeling, non linear and optimal control

Machine Learning: PyTorch, JAX, Numpy

Tools: Linux, Git, Docker, LaTeX, OpenCV, CVXPY, JuMP

Honors and awards

Lead The Future Mentorship *2023 - Present*

- 13% Admission rate: Mentorship programme for STEM Excellence in Italy

Best freshmen award *2022*

- 1000€ award given to a percentage of the top performing students in the first year of a Bachelor's degree program